

FurryOS Sovereign Universe - System Specification

Version: 8.1.0 **Codename:** Sovereign Universe **Date:** 2026-01-01 **Author:** Thomas B. Sweet (Anthro Teacher) **License:** MIT

Part 1: System Genome

1. Project Metadata

- **Framework Name:** FurryOS Genome
- **Initial Model:** Gemini 3 Pro (API)
- **Revision Model:** Claude 4.5 Sonnet (Perplexity Pro)
- **Provenance:**
 - Blockchain Anchor: Bitcoin Block 929481
 - Asset Source: anthroheart.com
- **Philosophy:** Minimal live installer, maximum user choice, bleeding-edge stability, user empowerment.

2. Key Features

- Wayland as preferred display server.
- Enhanced Btrfs integration with automatic snapshots and rollback.
- Optimized ZRAM with ZSTD compression.
- Comprehensive dynamic theming capabilities.
- Default PipeWire audio server.
- Integrated Flatpak support.
- Secure Boot and TPM 2.0 integration.
- Systemd-homed for secure/portable user directories.
- Advanced power management (TLP) & modern kernel features (PDS scheduler, BPF).
- Optional immutable-like root filesystem resilience.

3. Live Environment

- **Visual Indicator:** Animated pulsing border (#FF6B35) with message "LIVE MODE - NOT INSTALLED YET".
- **Capabilities:** Hardware testing, Wifi/Web access, Installer wizard, System diagnostics.
- **Requirements:** 512MB RAM minimum (2GB recommended).

4. Installer Specification (Hybrid Live-Net)

- **Strategy:** Offline (USB fast install) or Online (Download latest packages).
- **Wizard Steps:**
- **Welcome:** Select level (Granny/Beginner, Gamer/Intermediate, Hacker/Advanced, Ghost/Paranoid).
- **Hardware:** Auto-detect CPU/GPU/TPM. Prompts for proprietary drivers (Nvidia/AMD/Wifi).
- **Storage:**
 - **Filesystem:** Btrfs (Recommended) with layout @ @home @var @opt @srv @cache @log @tmp @swap.
 - **Encryption:** LUKS2 AES-256-XTS (Optional TPM/Yubikey unlock).

- **Root Strategy:** Mutable or Immutable (OverlayFS).
- **Packages:** Select Desktop (GNOME/KDE/MATE/Sway/Hyprland) and Bundles (Gaming, Dev, Server, etc.).
- **Network:** Hostname setup, UFW firewall, Optional VPN.
- **Users:** Root locked. Admin user setup. Optional systemd-homed encryption.

5. System Taxonomy

- **Base:** Debian 13 (Bookworm/Trixie context).
- **Kernel:** Mainline Linux 6.12+ (Custom patches: ZRAM, Realtime audio, PDS scheduler).
- **Bootloader:** GRUB2 (Default) or Systemd-boot (Alternative).
- **Security:** AppArmor (Enforcing), Secure Boot, TPM integration.
- **Package Management:** Apt (CLI), FurryOS Browser (GUI), Flatpak (Default), Snap (Optional).
- **Audio:** PipeWire.
- **Update Manager:** Atomic/Transactional with Btrfs rollback.

6. Build & Assets

- **ISO Target:** ~300MB (Net Installer) / 1.2GB (Live Core).
- **Base Image:** Debian Live 13.2.0 AMD64 MATE.
- **Included Assets:** Wallpapers, Icons, Wayland Configs, Fonts.
- **Python Version:** 3.12+ (Externally managed removed).

7. Known Pain Points & Resolutions

- **Nvidia:** Auto-detect logic with Wayland compatibility layers.
- **Wifi:** Firmware (iwlwifi/realtek) included.
- **Updates:** Atomic updates via snapshots to prevent breakage.
- **Legacy Apps:** XWayland enabled by default.

Part 2: User Configuration Profile

1. User Details

- **Username:** anthro_user
- **Full Name:** Anthro Architect
- **Timezone:** America/Chicago
- **Shell:** /bin/zsh (Supercharged)

2. Build Preferences

- **Target Hardware:** Generic x86_64
- **Kernel:** 6.12 Mainline
- **Experience Level:** Intermediate

3. Hardware Overrides (Supercharged)

- **Display Server: Wayland** (Explicitly Preferred)
- **Fractional Scaling:** Enabled
- **HDR Support:** Enabled
- **Variable Refresh Rate (VRR):** Enabled
- **Drivers:** Proprietary Wifi (On), Nvidia (Off).

4. Security

- **Firewall:** Enabled
- **Ad Blocking:** DNS Level Enabled
- **Telemetry:** Disabled
- **Disk Encryption:** Full Disk Encryption (LUKS)
- **AppArmor:** Enabled

5. Storage & Persistence

- **Persistence Partition:** 4GB
- **Filesystem:** Btrfs
- **Snapshots:** Enabled
- **Compression:** ZSTD
- **Layout:** Standard

6. System Optimizations

- **Memory:** ZRAM Enabled (50% Ratio)
- **Bootloader:** systemd-boot
- **DNS:** systemd-resolved
- **NTP:** chrony
- **Audio:** pipewire

7. Desktop Experience

- **Environment:** GNOME
- **Theme:** Adw-gtk3 (GTK), Adwaita (Icons/Cursor)
- **Typography:** Subpixel RGB rendering, Full hinting.

8. Software Selection

- **Bundles:**
 - Gaming: Enabled (Steam, Lutris)
 - Dev Tools: Enabled (VSCode, Git, Python)
 - Creative/Office: Disabled
- **Distribution:** Flatpak (Enabled), Podman (Enabled), Snap (Disabled).
- **Power:** TLP Enabled.

Appendix: PDF Generation Utility

Recommended Tool for Debian 13: Pandoc

Pandoc is the universal document converter and is standard in the Debian repositories. It produces high-quality PDFs via LaTeX.

Installation:

```
sudo apt update
sudo apt install pandoc texlive-latex-base texlive-fonts-recommended
```